# Crude Oil Inventory Event Strategy (WTI Term-Structure Proxy)

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## Abstract

This report presents an event-driven commodities research framework analyzing how a WTI term-structure proxy responds to weekly U.S. Energy Information Administration (EIA) crude inventory releases. Inventory surprises are standardized into signals and evaluated through a fixed-horizon backtest with transaction costs. Results indicate strong predictive power at the signal level, though naive execution fails to translate this into profitable performance.

## 1 Introduction

Weekly EIA crude oil inventory releases are a key macroeconomic indicator influencing energy markets. Unexpected inventory changes can signal supply-demand imbalances, potentially affecting the term structure of crude oil futures.

This study evaluates whether standardized inventory surprises provide predictive information for short-term movements in a WTI term-structure proxy.

## 2 Data

The analysis uses:

- Weekly U.S. crude oil inventory data from the EIA (2010–2026)
- Daily WTI futures-based spread proxy (2015–2026)

Inventory release dates are aligned to trading days to ensure accurate mapping between events and market response.

## 3 Signal Construction

Expected inventory is computed using a rolling average:

$$\text{Expected}_t = \frac{1}{N} \sum_{i=1}^{N} \text{Actual}_{t-i}$$

The inventory surprise is defined as:

$$\text{Surprise}_t = \text{Actual}_t - \text{Expected}_t$$

Signals are standardized using a rolling z-score:

$$Z_t = \frac{\text{Surprise}_t - \mu}{\sigma}$$

Trading rules:

- Long spread if $Z_t \leq -1$
- Short spread if $Z_t \geq 1$
- No position otherwise

# 4 Backtest Design

The strategy follows an event-driven framework:

- Fixed 5-day holding period
- Transaction costs applied
- Position clipping to prevent overlap

Daily PnL is computed as:

$$\text{PnL}_t = \text{Position}_{t-1} \cdot \Delta\text{Spread}_t - \text{Costs}$$

# 5 Results

## 5.1 Strategy Performance

The event-driven strategy produces a total return of -13.79 and an annualized return of -1.23, with a Sharpe ratio of -0.03. The strategy experiences a maximum drawdown of -110.45, indicating significant downside risk.

Despite a trade-level win rate of 51.28%, the daily win rate is only 19.39%, suggesting that trade timing and signal persistence are suboptimal. The average trade PnL is slightly negative (-0.0895), while the median trade is positive (0.1795), indicating that a small number of large losses dominate overall performance.

## 5.2 Event Study Analysis

The event study reveals a strong and persistent response of the spread proxy following inventory releases. Average spread changes increase from 0.099 at t+1 to 1.364 at t+5, with approximately 65% of events producing positive responses.

The peak response occurs at t+5, suggesting that the market reaction to inventory surprises is gradual rather than immediate.

## 5.3 Key Insight

The results highlight a divergence between signal strength and trading performance. While inventory surprises exhibit statistically meaningful predictive power, the execution framework fails to capture this alpha effectively.

# 6 Quantitative Metrics

## 6.1 Strategy Metrics

| Metric | Value |
|---|---|
| Total Return | -13.79 |
| Annualized Return | -1.23 |
| Sharpe Ratio | -0.03 |
| Max Drawdown | -110.45 |
| Daily Win Rate | 19.39% |
| Number of Trades | 78 |
| Trade Win Rate | 51.28% |
| Average Trade PnL | -0.0895 |
| Median Trade PnL | 0.1795 |
| Best Trade | 12.49 |
| Worst Trade | -15.41 |
| Average Holding Period | 14.23 days |

Table 1: Backtest Performance Metrics

## 6.2 Event Study Metrics

| Metric | Value |
|---|---|
| Avg Spread Change (t+1) | 0.099 |
| Avg Spread Change (t+3) | 1.295 |
| Avg Spread Change (t+5) | 1.364 |
| Positive Response Rate (t+1) | 65.60% |
| Positive Response Rate (t+3) | 65.48% |
| Positive Response Rate (t+5) | 64.54% |
| Peak Response Day | t+5 |
| Peak Magnitude | 1.364 |

Table 2: Event Study Results

# 7 Visualization

## 7.1 Cumulative PnL

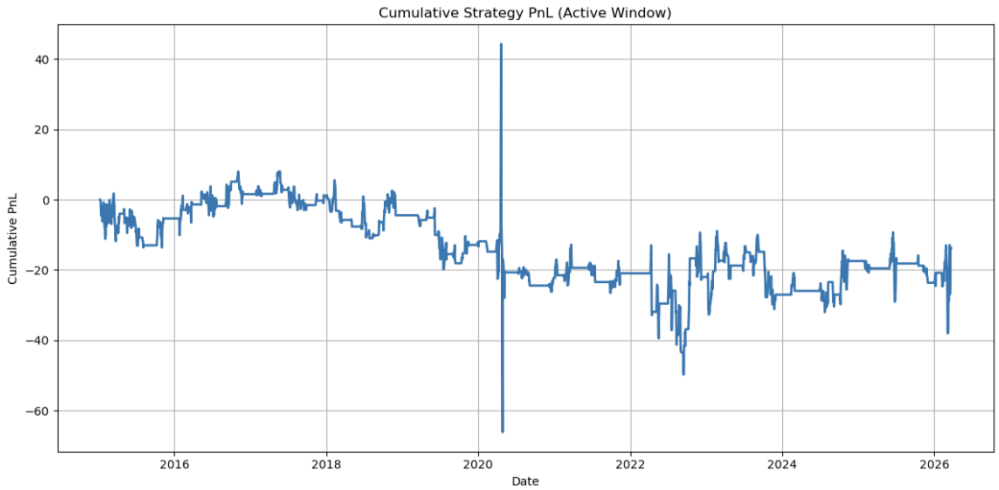


Figure 1: Cumulative strategy PnL

## 7.2 Strategy Position

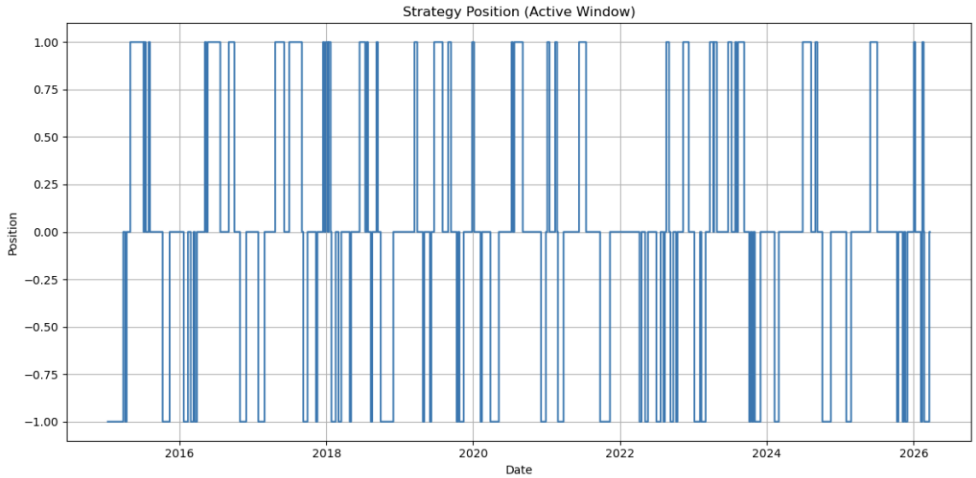


Figure 2: Strategy position over time

### 7.3 Spread with Signals

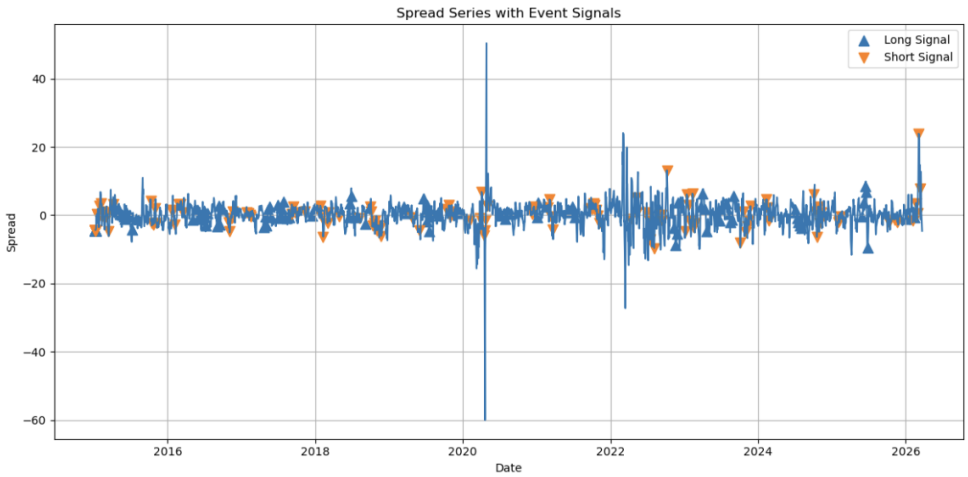


Figure 3: Spread series with long/short signals

### 7.4 Event Study

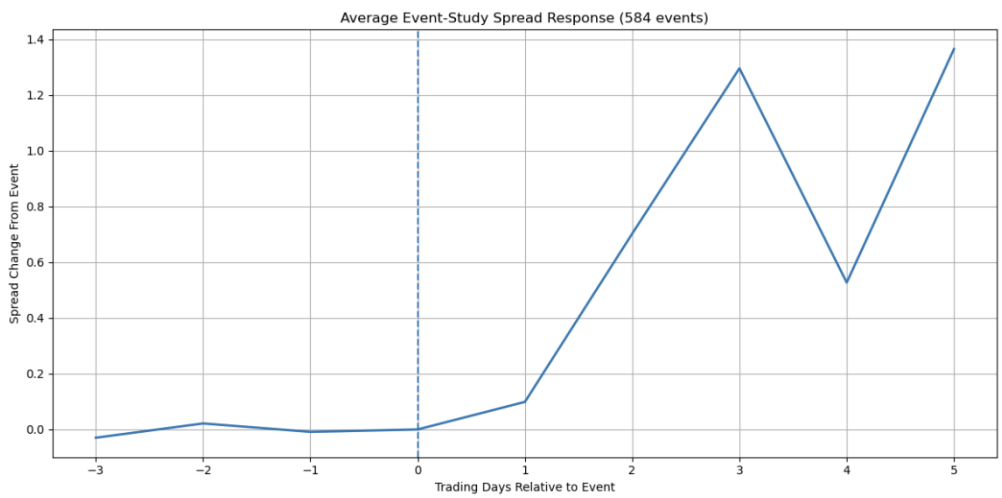


Figure 4: Average spread response around inventory releases

## 8 Limitations

- Spread is a proxy rather than true calendar spread (CL1–CL2)
- Execution does not account for slippage or liquidity constraints
- No regime filtering or volatility scaling

## 9 Future Work

- Use true futures calendar spreads (CL1–CL2)
- Optimize signal thresholds and holding periods
- Incorporate volatility targeting
- Apply machine learning for signal refinement

## 10 Conclusion

This study demonstrates that crude inventory surprises contain predictive information for short-term spread movements. While the signal is statistically meaningful, improved execution is required to translate predictive power into profitable strategies.